

Example 1: Campaign Response

Sample 1: $x_1=90$, $n_1=150$ ($\hat{p}_1=0.60$). Sample 2: $x_2=70$, $n_2=160$

($\hat{p}_2 = 0.4375$) . Test $\alpha = 0.05$ two-tailed .

- Pooled $\hat{p} = (90+70)/(150+160) = 0.518$.
- $z = (0.60 - 0.4375)/\sqrt{(0.518 \cdot 0.482 \cdot (1/150 + 1/160))} \approx 2.78$.
- Since $|2.78| > 1.960$, reject H_0 ; difference significant.
- 95% CI: $(0.60 - 0.4375) \pm 1.960 \cdot \sqrt{(0.60 \cdot 0.40/150 +$

$0.4375 \cdot 0.5625 / 160) \Rightarrow (0.05, 0.25)$.

Example 2: Treatment Success (Right-Tailed)

$x_1=45$, $n_1=60$ ($\hat{p}_1=0.75$). $x_2=38$, $n_2=65$ ($\hat{p}_2=0.585$). Test H_1 :

$p_1 > p_2$ at $\alpha = 0.01$.

- Pooled $\hat{p} = (45+38)/(60+65) \approx 0.664$.
- $z = (0.75 - 0.585)/\sqrt{(0.664 \cdot 0.336 \cdot (1/60 + 1/65))} \approx 2.33$.
- Critical $z = 2.326 \Rightarrow$ barely reject H_0 ($p \approx 0.010$).
- Conclusion: Treatment has higher success at the 1% level

(just significant) .